

Practical Week-10

1. Implement a file copy utility using low-level Linux file I/O system calls (open, read, write, lseek, close). Compare its performance with standard library functions (fopen, fread, fwrite).

```
imvarls@LAPTOP-VQ4D70L7: × + ∨

imvarls@LAPTOP-VQ4D70L7:~$ echo "Hello from Arun" > original.txt
imvarls@LAPTOP-VQ4D70L7:~$ cat original.txt
Hello from Arun
imvarls@LAPTOP-VQ4D70L7:~$ ls -i original.txt
53044 original.txt
imvarls@LAPTOP-VQ4D70L7:~$ stat original.txt
  File: original.txt
  size: 16          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 53044         Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/ imvarls)   Gid: ( 1000/ imvarls)
Access: 2026-09-25 10:37:51.215452869 +0000
Modify: 2026-09-25 10:37:36.555459604 +0000
Change: 2026-09-25 10:37:36.555459604 +0000
 Birth: 2026-09-25 10:37:36.555459604 +0000
imvarls@LAPTOP-VQ4D70L7:~$ ln original.txt hardlink.txt
imvarls@LAPTOP-VQ4D70L7:~$ ls -li original.txt hardlinnk.txt
ls: cannot access 'hardlinnk.txt': No such file or directory
53044 -rw-r--r-- 2 imvarls imvarls 16 Sep 25 10:37 original.txt
imvarls@LAPTOP-VQ4D70L7:~$ la -li original.txt hardlink.txt
53044 -rw-r--r-- 2 imvarls imvarls 16 Sep 25 10:37 hardlink.txt
53044 -rw-r--r-- 2 imvarls imvarls 16 Sep 25 10:37 original.txt
imvarls@LAPTOP-VQ4D70L7:~$ echo "Modified using hard link" >> hardlink.txt
imvarls@LAPTOP-VQ4D70L7:~$ cat original.txt
Hello from Arun
Modified using hard link
imvarls@LAPTOP-VQ4D70L7:~$ stat original.txt
  File: original.txt
  size: 41          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 53044         Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/ imvarls)   Gid: ( 1000/ imvarls)
Access: 2026-09-25 10:42:54.255332579 +0000
Modify: 2026-09-25 10:42:41.411337845 +0000
Change: 2026-09-25 10:42:41.411337845 +0000
 Birth: 2026-09-25 10:37:36.555459604 +0000
imvarls@LAPTOP-VQ4D70L7:~$ ln -s original.txt symlink.txt
imvarls@LAPTOP-VQ4D70L7:~$ ls -li original.txt hardlink.txt symlink.txt
53044 -rw-r--r-- 2 imvarls imvarls 41 Sep 25 10:42 hardlink.txt
53044 -rw-r--r-- 2 imvarls imvarls 41 Sep 25 10:42 original.txt
53029 lrwxrwxrwx 1 imvarls imvarls 12 Sep 25 10:43 symlink.txt -> original.txt
imvarls@LAPTOP-VQ4D70L7:~$ ls -i original.txt
53044 original.txt
imvarls@LAPTOP-VQ4D70L7:~$ find. -inum 53044
Command 'find.' not found, did you mean:
  command 'findv' from deb polylib-utils (5.22.5-4.2+dfsgbuild2)
  command 'find' from deb findutils (4.10.0-3build2)
  command 'findg' from deb ncl-ncarg (6.6.2.dfsg.1-12)
Try: sudo apt install <deb name>
imvarls@LAPTOP-VQ4D70L7:~$ find . -inum 53044
./hardlink.txt
./original.txt
imvarls@LAPTOP-VQ4D70L7:~$ rm original.txt
imvarls@LAPTOP-VQ4D70L7:~$ cat hardlink.txt
Hello from Arun
Modified using hard link
imvarls@LAPTOP-VQ4D70L7:~$ |
```

2. Develop a program that redirects standard input and output using file descriptors and `dup2()`. Demonstrate how shells implement I/O redirection.

```
imvarls@LAPTOP-VQ4D70L7: × + ▾
GNU nano 8.7.1 mmap_file.c *
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mmap.h>
#include <sys/stat.h>
#include <string.h>

int main() {
    int fd;
    struct stat st;
    char *data;

    fd = open("mmap_input.txt", O_RDWR);
    if (fd < 0) {
        perror("open");
        return 1;
    }

    if (fstat(fd, &st) < 0) {
        perror("fstat");
        close(fd);
        return 1;
    }

    data = mmap(NULL, st.st_size, PROT_READ | PROT_WRITE, MAP_SHARED, fd, 0);
    if (data == MAP_FAILED) {
        perror("mmap");
        close(fd);
        return 1;
    }

    printf("Original file content:\n%s\n", data);

    // Modify file content directly in memory
    strcpy(data, "Modified using mmap!");
    printf("File modified successfully.\n");

    msync(data, st.st_size, MS_SYNC);
    munmap(data, st.st_size);
    close(fd);

    return 0;
}
```

Output:

```
imvarls@LAPTOP-VQ4D70L7: × + ▾
imvarls@LAPTOP-VQ4D70L7:~$ echo "Hello from mmap! This file will be modified." > mmap_input.txt
imvarls@LAPTOP-VQ4D70L7:~$ cat mmap_input.txt
cat: mmap_input.txt: No such file or directory
imvarls@LAPTOP-VQ4D70L7:~$ echo "Hello from mmap! This file will be modified." > mmap_input.txt
imvarls@LAPTOP-VQ4D70L7:~$ cat mmap_input.txt
Hello from mmap! This file will be modified.
imvarls@LAPTOP-VQ4D70L7:~$ nano mmap_file.c
imvarls@LAPTOP-VQ4D70L7:~$ gcc mmap_file.c -o mmap_file
imvarls@LAPTOP-VQ4D70L7:~$ ./mmap_file
Original file content:
Hello from mmap! This file will be modified.

File modified successfully.
imvarls@LAPTOP-VQ4D70L7:~$ |
```